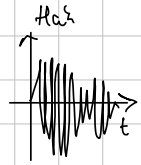
6.1. SUSTAVI

~Uvod~

ulaz (signali) → SUSTAV → izlaz (signali) ⇒



MIKROFON

odnosno:

$u(t)$, $u[n]$, $u \in U$ → $S: U \to Y$ → $y(t)$, $y[n]$, $y \in Y$

skup svih izlaznih signala

skup svih ulaznih signala

Primjer: (kont. signali)

signali su funkcije vremena

$u(t) = \sin(2t + \frac{\pi}{3})$ → $y(t) = 3 \cdot u(t)$ → $y(t) = 3 \cdot \sin(2t + \frac{\pi}{3})$

sustav nije fja od t

⇒ izlaz je funkcija od ulaza

- ulazni i izlazni signali kontinuirani signali ⇒ KONTINUIRANI SUSTAV (t)
- Disk. SUSTAV = Disk. signal → Disk. signal $[n]$

∀ kris kas komlo nešćemo raditi

Primjer: Disk. signal

$$y[n] = \frac{1}{L+1} \sum_{m=0}^{L} u(n-m)$$

SUSTAV USREDNJAVANJA

$$y[n] = \frac{1}{4} \sum_{m=0}^{3} u[n-m]$$

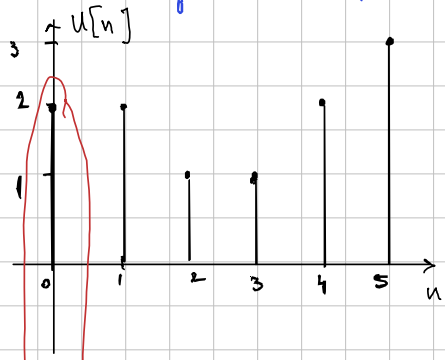
$$= \frac{1}{4} \left(u[n-0] + u[n-1] + u[n-2] + u[n-3] \right)$$

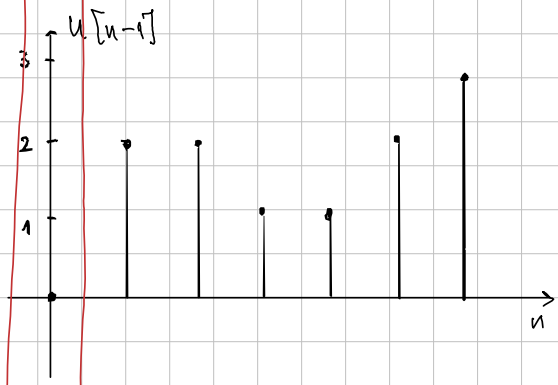
pomaknut za 1 mj. — pom. 2 mjesta — pom. 3 mjesta

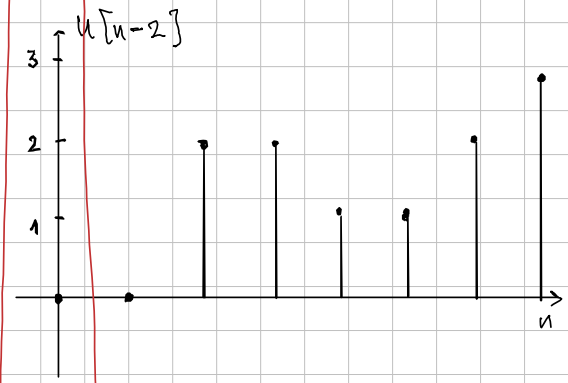
- uzeli smo 4 susjedne vr. nekog signala, zbrojili i podijelili s 4 ⇒ USREDNJILI

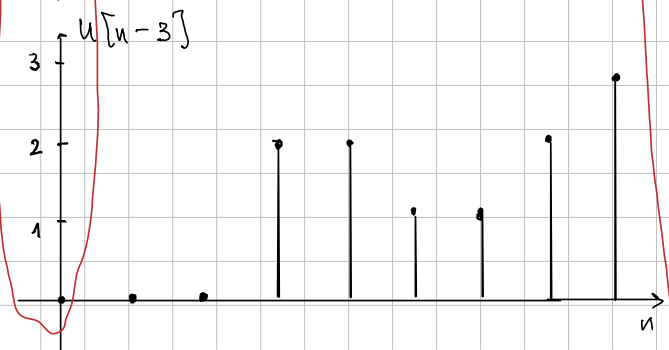
Primjer: Diskretni

SUSTAV USREDNJAVANJA

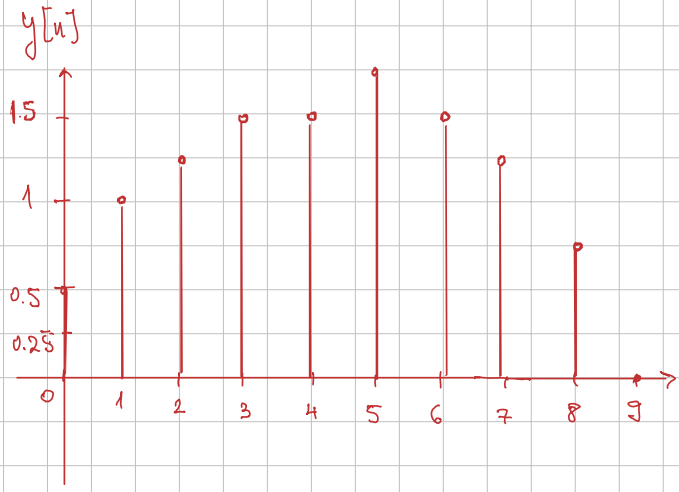








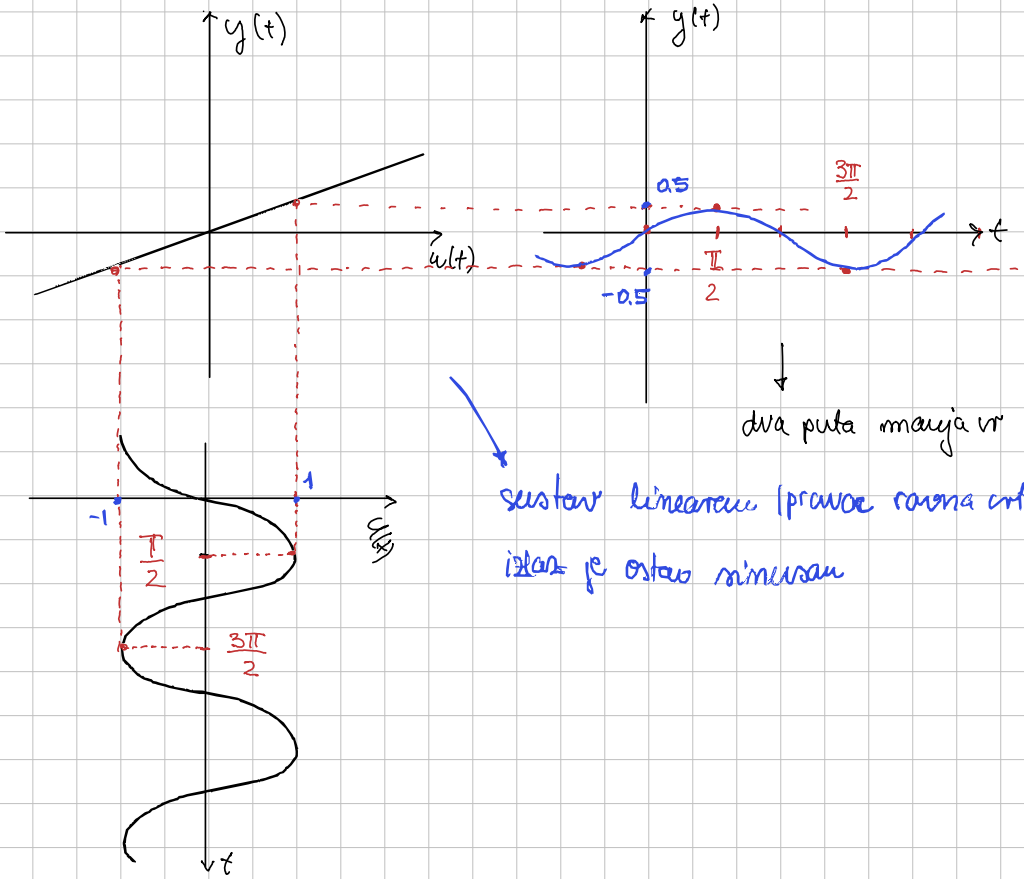
Zbrojimo ulaze i podijelimo s 4



Primjer: Vremenski kontinuirani

LINEARNI SUSTAV

$y(t) = 0.5\, u(t)$

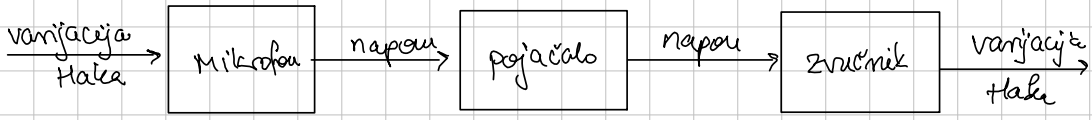


sustav linearan (pravac ravna crta)

izlaz je ostao sinusan

Povezivanje sustava

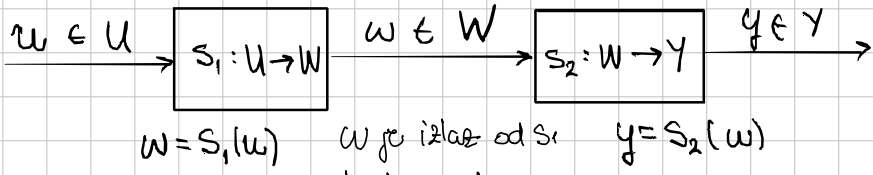
· isti ulaz može biti modeliran kroz više različitih sustava



↳ ovo je kaskadni spoj

postoje: kaskadni, paralelni i spoj sustava u povratnu vezu

1. Kaskada:



$w = S_1(u)$ — w je izlaz od S_1 i ulaz od S_2 — $y = S_2(w)$

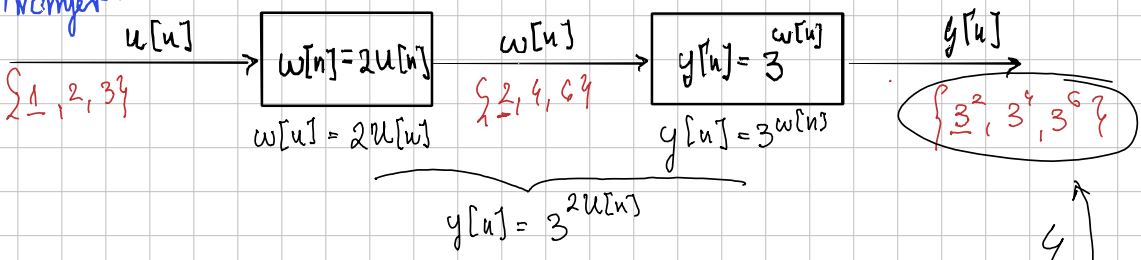
$w(t) = S_1(u(t))$ — $y(t) = S_2(w(t))$

$w[n] = S_1[u[n]]$ — $y[u] = S_2[w[n]]$

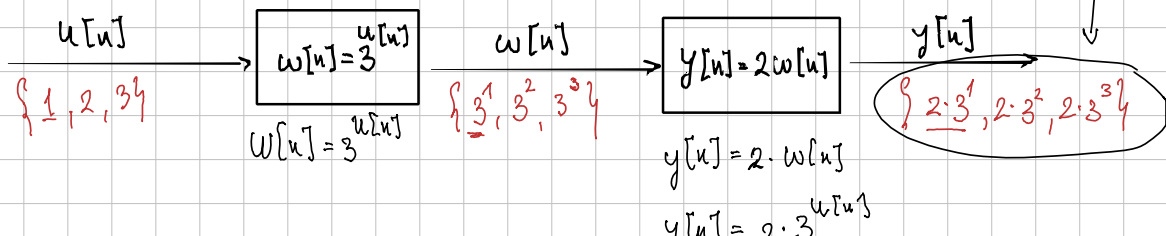
$y(t) = S_2(S_1(u(t)))$

(kompozicija fja →) $y[u] = S_2[S_1[u[n]]]$ $\Rightarrow S_2 \circ S_1$

Primjer:



$\neq$



! Kompozicija funkcije nije komutativna !

→ smijemo zamijeniti ako su sustavi međusobno inverzni

Inverzni sustav:

$$u \longrightarrow \boxed{S} \xrightarrow{w} \boxed{S^{-1}} \xrightarrow{y}$$

$$y = u$$

Primjer:

$$u \longrightarrow \boxed{w[n] = 2u[n]+1} \xrightarrow{w} \boxed{S^{-1}} \xrightarrow{y}$$

da bi dobili inverz moramo

$$w = 2u+1$$

$$u = 2w+1$$

$$w-1 = 2w$$

$$w = \frac{u-1}{2}$$

$$y = \frac{w-1}{2}$$

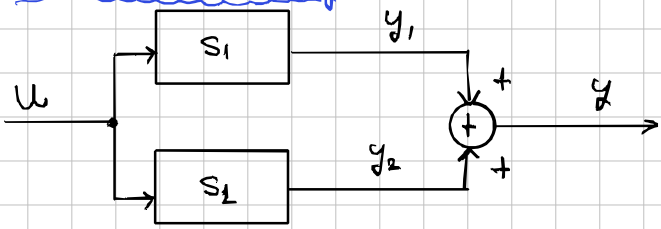
$$y = \frac{2w+1-1}{2} \Rightarrow y = u$$

$$u \longrightarrow \boxed{w = \frac{u-1}{2}} \xrightarrow{w} \boxed{y = 2w+1} \xrightarrow{y}$$

$$\hookrightarrow y = \frac{2w+1}{2} = \frac{2(u-1)+1}{2} \Rightarrow y = u$$

→ Samo u kaskadi međusobno inverznih sustava je primjenjiva komutat.

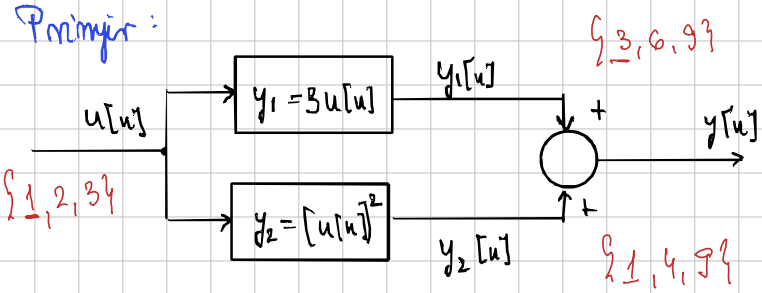
2. Paralelni spoj



$y = y_1 + y_2$

$y_1 = S_1(u) \qquad y_2 = S_2(u)$

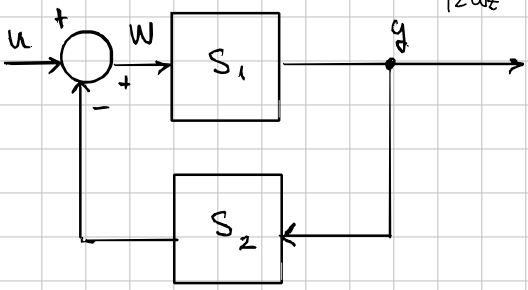
Primjer:



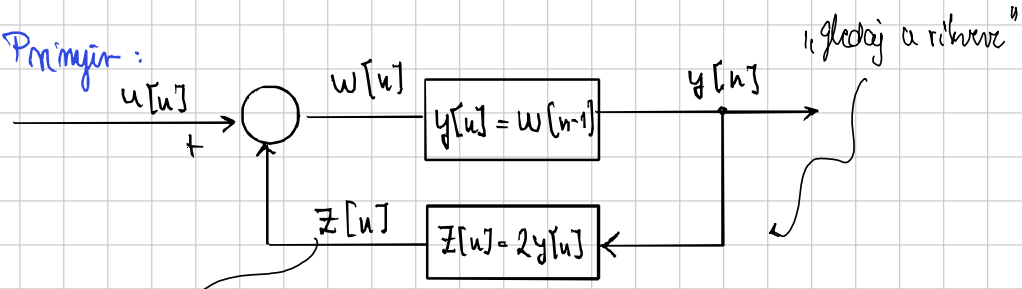
$y[n] = y_1[n] + y_2[n] = 3u[n] + (u[n])^2$

3. Povratna veza → moramo znati izlaz da bismo izračunali izlaz

izlaz vodimo na ulaz da dobijemo izlaz



Primjer:



$z[n] = 2y[n]$

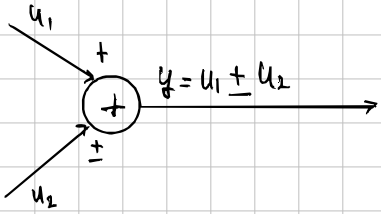
$w[n] = u[n] + 2y[n]$

$y[n] = w[n-1] = u[n-1] + 2y[n-1]$

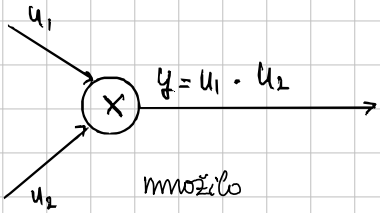
! izlaz u nekom trenutku ovisi o ulazu u prošlom trenutku i izlazu u prošlom trenutku!

PRIKAZ SUSTAVA BLOKOVSKIM DIJAGRAMOM

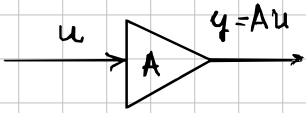
Osnovni blokovi



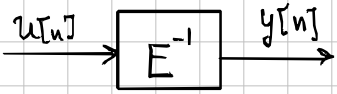
Zbrajalo s 2 ili više ulaza



množilo



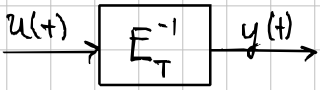
množilo s konstantom



$y[n] = u[n-1]$

$\forall n \in \mathbb{Z}$

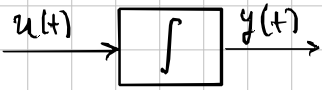
element za jediničnu kašnjenje



$y(t) = u(t-T)$

$\forall t \in \mathbb{R}, T \in \mathbb{R}_+$

element za kašnjenje



$$y(t) = \int_{-\infty}^{t} u(\lambda)\,d\lambda$$

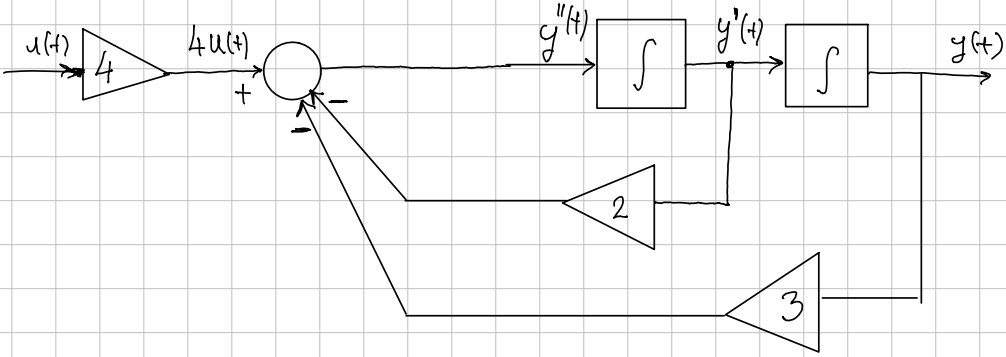
integrator

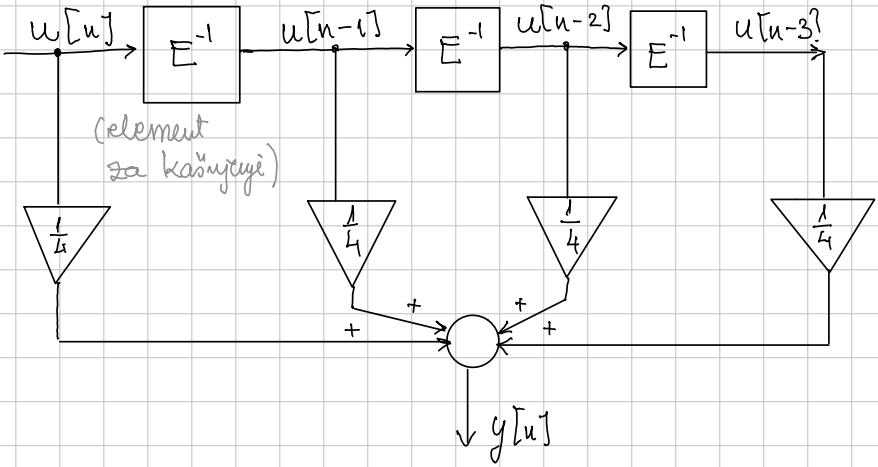
Primjer: električna mreža (glazbena utica)

$$y''(t) + \frac{R}{L} y'(t) + \frac{1}{LC} y(t) = \frac{1}{LC} u(t)$$

$$y''(t) + 2y'(t) + 3y(t) = 4u(t)$$

$$y''(t) = 4u(t) - 2y'(t) - 3y(t)$$



Primjer: Diskretni sustav:

$$y[n] = \frac{1}{4}u[n] + \frac{1}{4}u[n-1] + \frac{1}{4}u[n-2] + \frac{1}{4}u[n-3]$$